

$$\frac{ghm \cos (\alpha(t))}{2} + 0.5m \left(\frac{h^2 \sin^2 (\alpha(t)) \frac{d^2}{dt^2} \alpha(t)}{2} + \frac{h^2 \cos^2 (\alpha(t)) \frac{d^2}{dt^2} \alpha(t)}{2} \right) = -b \frac{d}{dt} \alpha(t) + \tau$$